Reinsurance Allocation Optimization

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March 1, 2026

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1 Introduction

Whether it be natural or man made, disasters pose a large risk for both survivors and insurers. Reinsurance contracts are a way for insurance companies to diminish risk and losses. It is a tool to share the weight and risk exposure of a possible catastrophe.

As important as it is, there are many variables to be taken into account when making a reinsurance contract. Maximizing profit while minimizing earnings volatility, adhering to standarts and regulations and the non-linear nature of layered structures makes the optimization a computational challenge. For this project we are following the work of G. Woodman et al. [1] and implementing QAOA to demonstrate another quantum approach for this problem.

2 Formulation and Mathematical Model

2.1 Problem and Assumptions

Following the literature, we are inspecting a treaty excess of loss scheme, and testing the algorithm with a randomly generated NatCat database. As typical of such a contract, we are working with a layered system, compromising the subgroup degree of freedom for the sake of simplicity and time limitation. Also assuming tower architecture, where there are no gaps between layers allows us to reduce another degree of freedom.

2.2 Decision Variables

In interest of profit optimization of the reinsurance contract, we can look at the decision variables in two parts.

2.2.1 Perils

Groups, g: Main coverage units, loss that adds up to the same Limit.
Subgroups, s: perils in the same group that share the same attachment point.

In the scope of this projects all perils are treated as if they appear with equal probability.

2.2.2 Layers

For each group, the contract may include multiple layers. These layers are characterized by,
Attachment point, A: Point where the reinsurance coverage of the layer starts.
Subgroup attachment shift, Δ_s: difference between the attachment points of different perils. Often shifted due to the severity and frequency differences of perils.
Layer limit, L: Range of the reinsurance coverage
Reinstallments, r: How many times the limit had to be reset. Since we are treating each peril equally, we will take $\Delta_s = 0$.

2.3 Objective Function

In this project, we aim to maximize the profits by optimizing A, L, and r per layer. Thus, our objective function is given by,

$$O(C) = avg_t(\pi_n et, t(C)) + \sum \lambda_K(C) \tag{1}$$

with λ_K denoting the constraints we define later.

The path from the parameters A_l,L_l and r_l to π_t is as follows: Recovery, the amount paid back by the reinsurer per event with loss $\mathcal{L}_l$, is given by,

$$R_{l,e} = min(max[0, \mathcal{L}_e - A_l, L_l]) \tag{2}$$

Total recovery per trial year t in a layer l is the sum of all per-event recoveries up to the limit L with r reinstallments is given by,

$$R_{t,l} = min((1 + r_l)L_l, \sum R_{l,e}) \tag{3}$$

Thus, giving average recovery,

$$\bar{R}_l = \frac{1}{|\{t\}|} \sum R_{t,l} \tag{4}$$

Retained loss is calculated by subtracting the recovery from all layers from gross event losses,

$$\mathcal{R}_t = \sum \mathcal{L}_e - \sum R_{t,l} \tag{5}$$

Price Estimation

The cost of signing this reinsurance contract is also something that should be taken into account. However estimating this price is a challenging task in and of itself. Without going into this rabbit hole we will define the reinsurance premium as,

$$\Pi_l^{rein} = (1 + \rho)\bar{R}_l \tag{6}$$

where ρ is an additional safety loading factor, which describes reinsurance market conditions. In order to further simplify this process we will replace it with a placeholder constant.

Reinstatement Premium

Assuming the price for reinstatement is the same as the original premium we will define reinstatement cost as,

$$\Pi_{t,l}^{rstm} = \Pi_l^{rein} \cdot \frac{min(r_l \cdot L_l, R_{t,l})}{L_l} \tag{7}$$

Then we can define net loss,

$$\mathcal{N}_t = \sum \mathcal{L}_e - \sum [R_{t,l} - \Pi_{t,l}^{rstm}] \tag{8}$$

This brings us to the net profit definition,

$$\pi_t = \Pi_{gross} - \mathcal{N}_t - \sum \Pi_l^{rein} \tag{9}$$

2.4 Practical Constraints

Risk

Purpose of the reinsurance is to reduce risk on the insurer. In this section we are directed by the literature to use Tail Value at Risk as the risk measure. Which give us the average of the yearly net loss values that are above a certain percentile which is, in TVaR case, typically 99.5%.

$$TVaR_{\beta}[\mathcal{N}_t] = |\mathcal{N}_t| : \mathcal{N}_t > \boldsymbol{P}_{\beta}(\mathcal{N}_t) \tag{10}$$

Constraints come from regulatory or business imposed limits which can be mathematically defined.

2.4.1 Aggregate Exceedance Probability

AEP measures and limits the total loss accumulated over all events in a trial year.

$$AEP_{\beta} = \boldsymbol{P}_{\beta}(\mathcal{N}_t) < AEP_{\beta}^{alert} \tag{11}$$

2.4.2 Occurrence Exceedance Probability

OEP focuses on single event severity. In this case, we select the largest loss event in a given trial year and apply recovery to its loss,
$\mathcal{L}_{p,t}^{max} = max(\mathcal{L}_e - \sum R_{l,e})$ which gives us,

$$OEP_{\beta,p} = \boldsymbol{P}_{\beta}(\mathcal{L}_{p,t}^{max}) < OEP_{\beta,p}^{alert} \tag{12}$$

2.4.3 Probability of attaching

This simply gives the probability that there will be a non-zero recovery in a given layer.

$$P_l^{att} = \frac{|\{R_{t,l} > 0\}|}{|\{t\}|} < P_l^{att,alert} \tag{13}$$

In the objective function these constraints contribute as,

$$\lambda_K = -\gamma_K \cdot ReLU(K - \tau_K) \tag{14}$$

where γ_K is the linear scaling constant and τ_K is the limit for constraints K.

2.5 Dataset

We generated two datasets randomly in Python. We used lognormal distribution to generate losses and Poisson distribution (with average of three events per year) for a sample time of 2000 years. In this case, lognormal distribution captures the rare (but still possible) nature of catastrophes. For the second data set, we used random indices to denote three different groups.

2.6 Complexity Analysis

We tried classically modeling this data using Simulated Annealing. It is mentioned in a review by H. Brinch [2] that this method has time complexity,

$$T_n = O((n^2 + n)\log n) \tag{15}$$

Quantum case:
G : number of groups
N : number of layers per group
$|L|$: size of limit grid
$|R|$: size of reinstatement grid
$|U| = |L||R|$: number of layer options
B_w : beam width in DP (number of partial towers kept)
E : total number of events (sum over groups), $E = \sum_g E_g$
n : number of binary variables (qubits) in portfolio QUBO
p : QAOA depth (reps)
I : number of classical optimizer iterations
S : number of objective evaluations per optimizer iteration (constant-factor, depends on optimizer)
shots : number of measurement shots per circuit (hardware / shot-based simulation)

$$T_{\mathrm{DP}} \;=\; \mathcal{O}\left(\sum_{g=1}^{G} N^2\, B_w\, |U|\, E_g\right) \;=\; \mathcal{O}\big(N^2\, B_w\, |U|\, E\big)\,.$$

$$M_{\mathrm{sim}} \;=\; \mathcal{O}(2^n)\,.$$

$$T_{\mathrm{QAOA,\,sim}} \;=\; \mathcal{O}(I\, S\, p\, n\, 2^n)\,.$$

$$M_{\mathrm{hw}} \;\approx\; \mathcal{O}(n) \quad \text{(qubits, ignoring classical logs).}$$

$$T_{\mathrm{QAOA,\,hw}} \;=\; \mathcal{O}(I\, S\, p\, \text{shots}) \quad \text{(up to a hardware-dependent constant per shot).}$$

$$T_{\mathrm{total,\,sim}} \;=\; \mathcal{O}\big(N^2\, B_w\, |U|\, E \;+\; I\, S\, p\, n\, 2^n\big)\,.$$

$$M_{\mathrm{total,\,sim}} \;=\; \mathcal{O}(E \;+\; 2^n)\,.$$

$$T_{\mathrm{total,\,hw}} \;=\; \mathcal{O}\big(N^2\, B_w\, |U|\, E \;+\; I\, S\, p\, \text{shots}\big)\,.$$

3 Quantum Algorithm

In the light of the parameters discussed above, it is now clearer to see how deep of a problem this is. So far we have assumed that there are no subgroups, all perils are equally likely and that there are no gaps between layers. The last assumption allows us to model the array of attachment points A_k as

$$A_{k+1} = A_k + L_k$$

then we only need one base Attachment point A_0 and the rest can me denoted as sum of limits added to A_0.

For this project we chose to implement a hybrid algorithm using Dynamic Programing for local optimization within one group and QAOA for global optimization to pick the best contract across multiple perils.

Furthermore, using Dynamic Programing along with our assumptions allows us to take a global optimization problem and turn it into a decision problem. This is particularly useful because considering multiple groups in one QUBO scheme requires too many qubits thus, exponentially increasing the memory complexity of the algorithm.

As a result of our quantum algorithm, we obtain the output in Fig. (1).

On the top section of this output, it is visible that Dynamic Programming works as a decision mechanism and provides us the appropriate contracts for certain events. The lower half of the output shows QAOA algorithm chooses the best fitted option for each group and returns the optimized parameters. The results are as follows:

Baseline tail (no reinsurance) = 597.50M
Selected tail proxy = 484.02M
Tail reduction = 113.48M (18.99%)
Total premium = 20.97M (budget 20.00M)
Tail reduction per premium = 5.41 (M tail reduced per M premium)

```
Loaded events: 5956, years: 1882, groups: ['0', '1', '2']
Loss scale in millions: min=0.0169, median=7.45, max=957
Tail years count = 19 (alpha=0.99); example tail years (first 10): [109, 1953, 135, 494, 817, 1360, 699, 1974, 535, 782]
DP: depth=1/4, partial towers kept = 12
DP: depth=2/4, partial towers kept = 30
DP: depth=3/4, partial towers kept = 30
DP: depth=4/4, partial towers kept = 30

Group 0: candidates kept = 2
  cand 0: premium=6.93M, cost=12.02M, layers=(L=25M,r=2), (L=25M,r=0), (L=25M,r=0), (L=25M,r=0)
  cand 1: premium=6.99M, cost=12.02M, layers=(L=25M,r=2), (L=25M,r=2), (L=25M,r=0), (L=25M,r=0)
DP: depth=1/4, partial towers kept = 12
DP: depth=2/4, partial towers kept = 30
DP: depth=3/4, partial towers kept = 30
DP: depth=4/4, partial towers kept = 30

Group 1: candidates kept = 2
  cand 0: premium=7.17M, cost=12.22M, layers=(L=25M,r=2), (L=25M,r=2), (L=25M,r=2), (L=25M,r=2)
  cand 1: premium=7.17M, cost=12.22M, layers=(L=25M,r=2), (L=25M,r=2), (L=25M,r=2), (L=25M,r=1)
DP: depth=1/4, partial towers kept = 12
DP: depth=2/4, partial towers kept = 30
DP: depth=3/4, partial towers kept = 30
DP: depth=4/4, partial towers kept = 30

Group 2: candidates kept = 2
  cand 0: premium=6.88M, cost=11.74M, layers=(L=25M,r=2), (L=25M,r=2), (L=25M,r=1), (L=25M,r=1)
  cand 1: premium=6.88M, cost=11.74M, layers=(L=25M,r=2), (L=25M,r=2), (L=25M,r=1), (L=25M,r=2)
2026-03-01 10:16:23.250 Python[33235:10654908] ApplePersistenceIgnoreState: Existing state will not be touched. New state

Portfolio QUBO (conceptual):
  minimize  Σ cost[g,m]*x[g,m] + W_tail*mean_tail_years(NetLoss_portfolio)
            + P_onehot*Σ_g (Σ_m x[g,m] - 1)^2
            + P_budget*(Σ premium[g,m]*x[g,m] - Budget)^2
  Budget=20.00M, alpha=0.99, tail_years=19, vars=6
  NOTE: tail-risk is approximated using a fixed tail-year set (scales to thousands of years).

Selected portfolio towers:

Group 0 (chosen cand 0): premium=6.93M, cost=12.02M
  layer 0: A=50M, L=25M, r=2
  layer 1: A=75M, L=25M, r=0
  layer 2: A=100M, L=25M, r=0
  layer 3: A=125M, L=25M, r=0

Group 1 (chosen cand 0): premium=7.17M, cost=12.22M
  layer 0: A=50M, L=25M, r=2
  layer 1: A=75M, L=25M, r=2
  layer 2: A=100M, L=25M, r=2
  layer 3: A=125M, L=25M, r=2

Group 2 (chosen cand 0): premium=6.88M, cost=11.74M
  layer 0: A=50M, L=25M, r=2
  layer 1: A=75M, L=25M, r=2
  layer 2: A=100M, L=25M, r=1
  layer 3: A=125M, L=25M, r=1
Tail-loss proxy (mean net loss on tail years) = 484.02M
QUBO objective value = -2.98791e+10

Baseline tail (no reinsurance) = 597.50M
Selected tail proxy            = 484.02M
Tail reduction                 = 113.48M (18.99%)
Total premium                  = 20.97M (budget 20.00M)
Tail reduction per premium     = 5.41 (M tail reduced per M premium)
```

Figure 1: Quantum Algorithm output.

For this algorithm the constraints we have considered are TVaR and an independent parameter. AEP and OEP are things we have once again cut out for time.

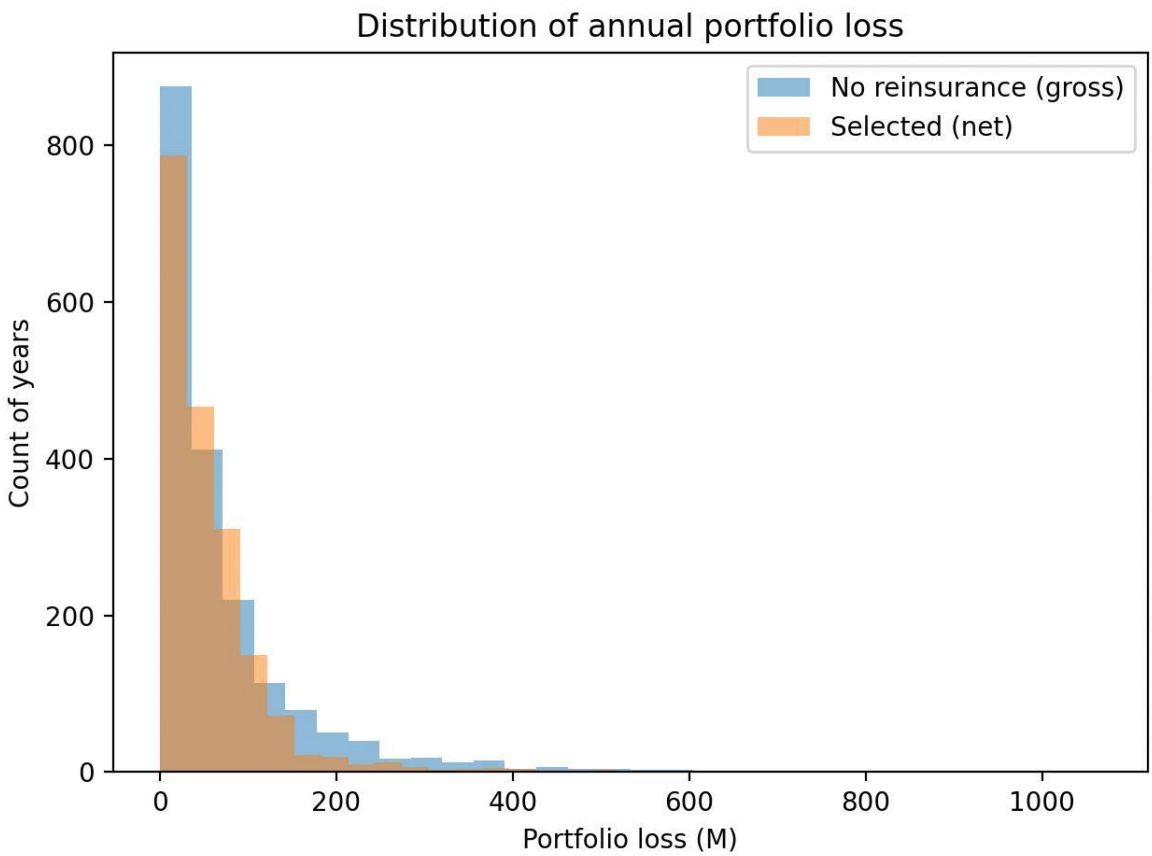

Figure 2: Distribution of Annual Portfolio Loss

4 Extra

For the extra section we chose to apply a classical benchmark using simulated annealing. Simulated annealing works by randomly exploring neighboring solutions and, occasionally accepting worse ones with a probability that decreases over time. This allows the estimation to escape local optima and converge towards a global optimum.

Same assumptions as above apply to this process, in addition we pick a constant scaling factor for constraints to make it easier to run. Due to time limit, we could only consider one constraint which is AEP. Maximization of objective function is realized by optimizing layer height and number of reinstatements. Scaling prefactor and threshold in the plots below denote γ_K and τ_K from equation (14).

As shown in Fig.(3), the objective function with reinsurance becomes much larger than the case without reinsurance as the scaling prefactor gamma increases. The Fig.(4) shows that the gap between objective function with reinsurance and without reinsurance decreases with as threshold C increases. This illustrates the hyperparameter sensitivity of this classical simulated annealing method for optimization.

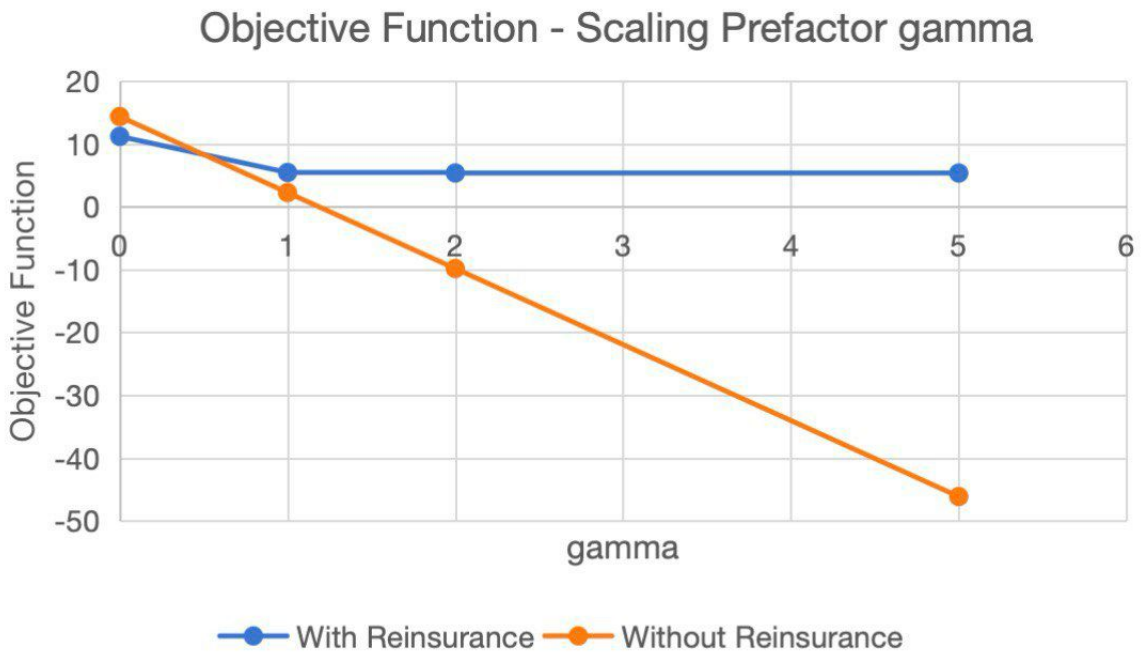

Figure 3: Objective Function vs Scaling Prefactor Gamma

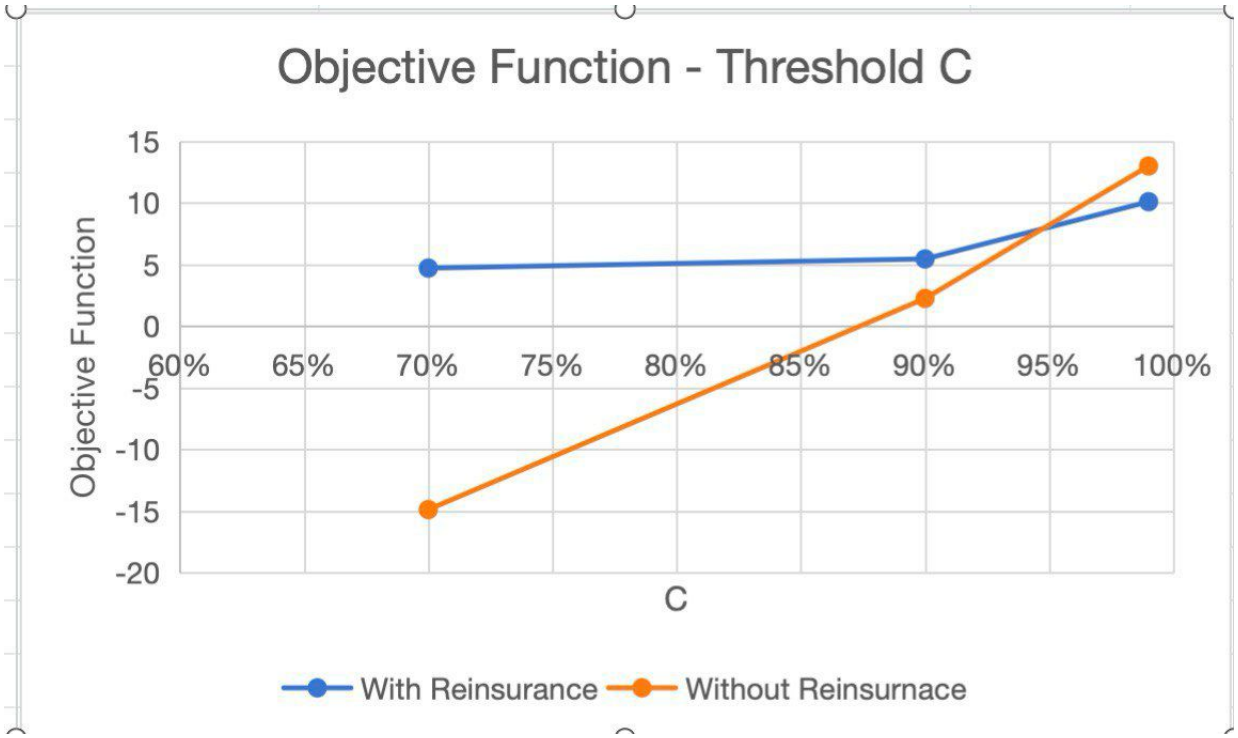

Figure 4: Objective Function vs Threshold C

References

[1] Woodman, G., Andrist, R. S., Häner, T., Steiger, D. S., Schuetz, M. J., Katzgraber, H. G., & Detyniecki, M. (2025). Modern Computational Methods in Reinsurance Optimization: From Simulated Annealing to Quantum Branch & Bound. arXiv preprint arXiv:2504.16530.

[2] Brinch, H. P. (1992). Simulated Annealing. Electrical Engineering and Computer Science-Technical Reports.